E. Generative Modeling

E.1 Generative Modeling: Basic concepts

E.2 Generative Modeling: Variational models

E.3 Generative Modeling: Diffusion models I

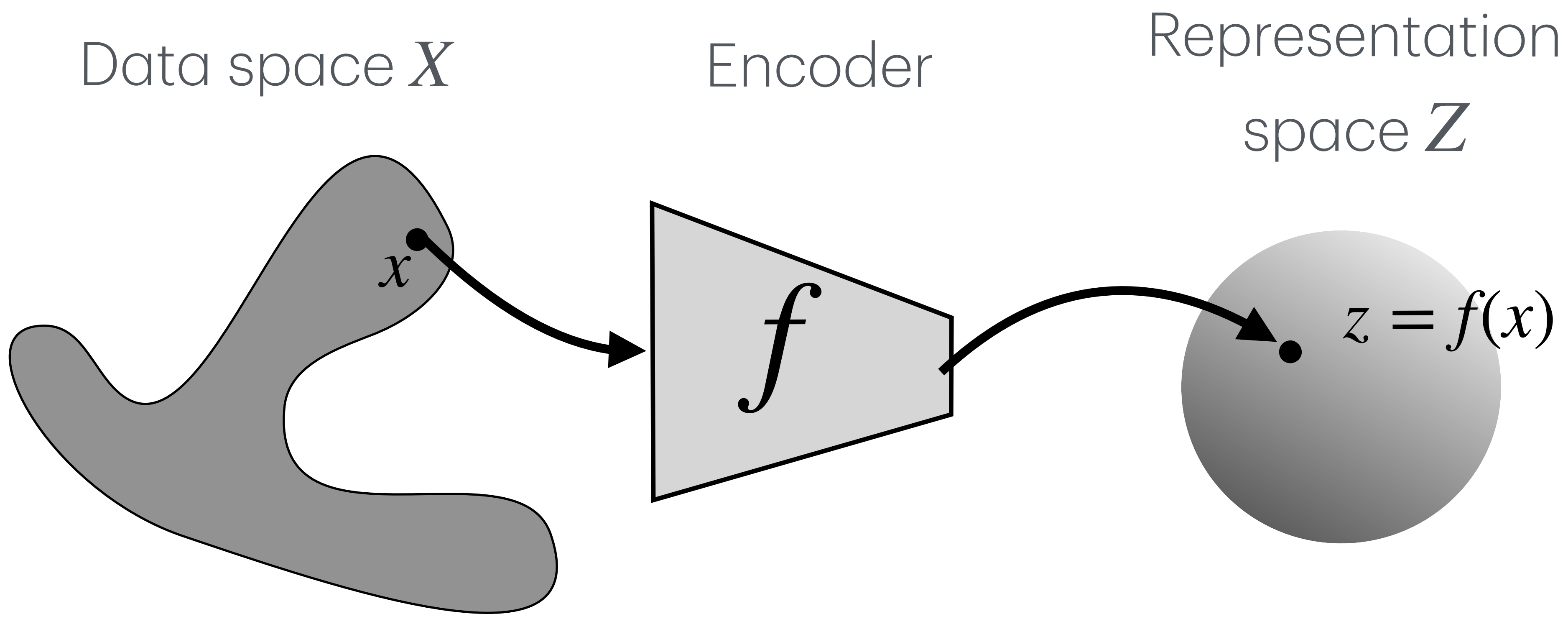
E.4 Generative Modeling: Diffusion models II

E.5 Generative Modeling: Structured (conditional) generation

Tiam Heydari
Postdoctoral Fellow, UBC

What is generative modeling?

Representation Learning



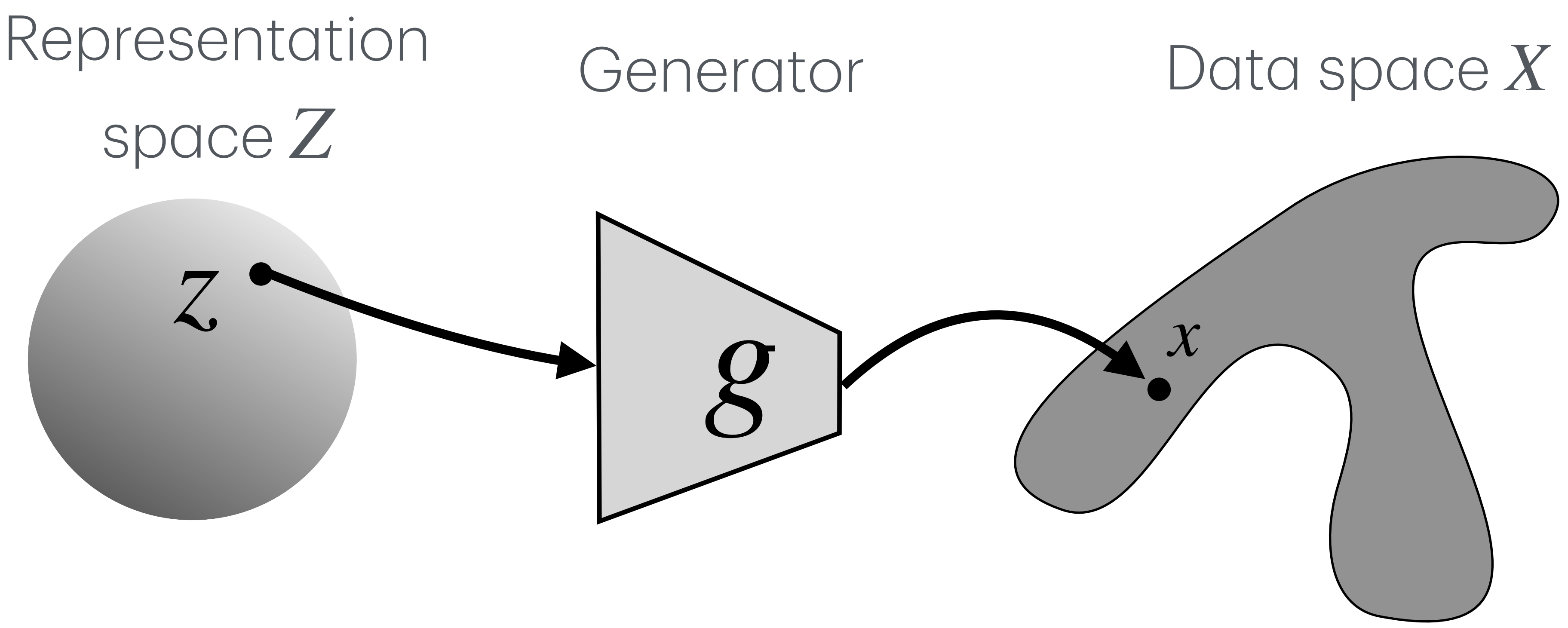
$$f : \mathcal{X} \to Z$$
$$z = f(x)$$

Question: Given data, what representation describes it?

Main direction: data → latent representation

Goal: compress, organize, compare, or predict from data.

Generative Modeling



$$g : Z \to \mathcal{X}$$
$$x = g(z)$$

Question: Given a latent code, what data could it produce?

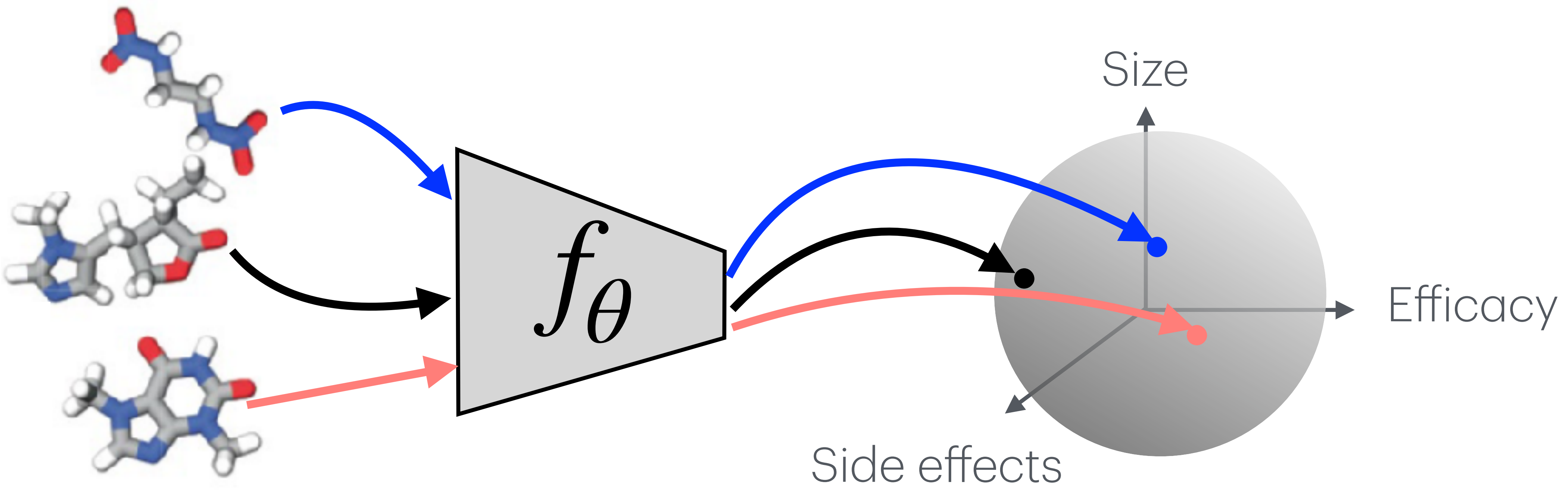
Main direction: latent representation → data

Goal: generate realistic new samples based on some variables

What is generative modeling?

Representation Learning

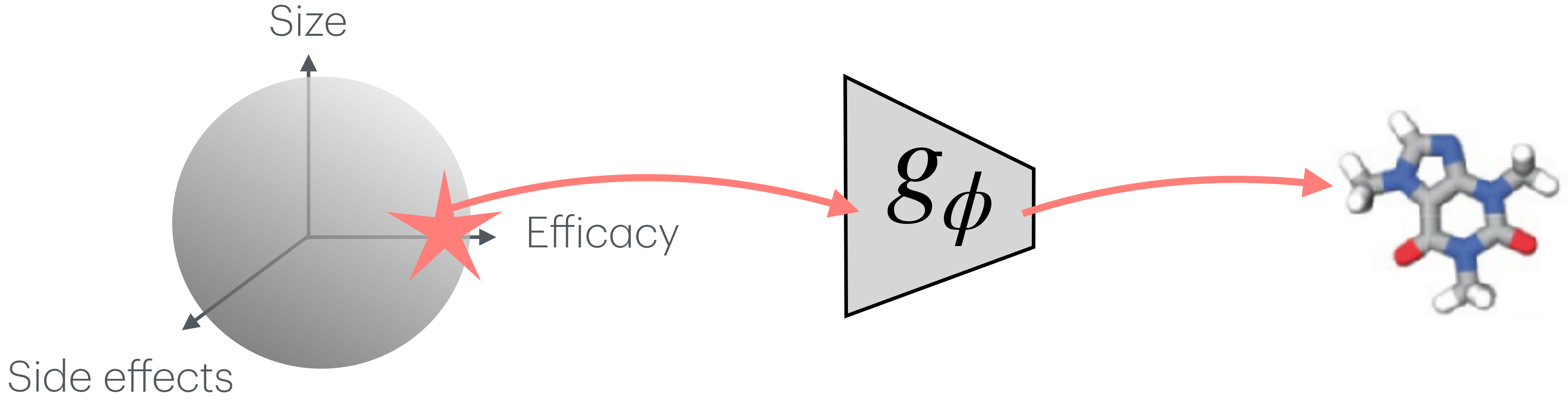
- For example: A successful Representation Learning for a drug molecular graphs we find a good disentangled latent space.



Now, if someone develops a new candidate structure x_{new}, we can use $f_\theta(x_{new})$ to compare it with other drugs in the database.

Generative Modeling

- For example: With a successful Generative Model for a drug molecular graphs we ask what is the drug structure with most efficacy and least side effects.



We just need to define which charachteristics in latent space we want z_{new}, and $x_{new} = g_\phi(z_{new})$ generates it *de novo!*

Wang, Hanchen, et al. "Scientific discovery in the age of artificial intelligence." *Nature* 620.7972 (2023): 47-60.

A historical intuition: biology as representation learning

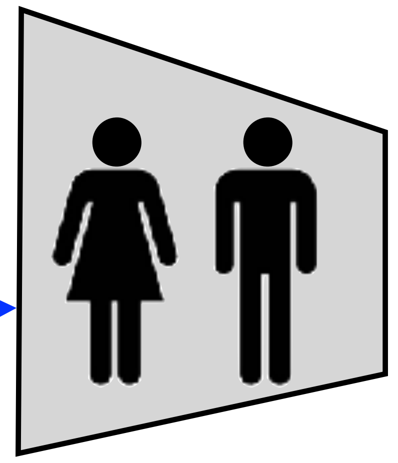
Living systems have many components, dimensions and are complex.

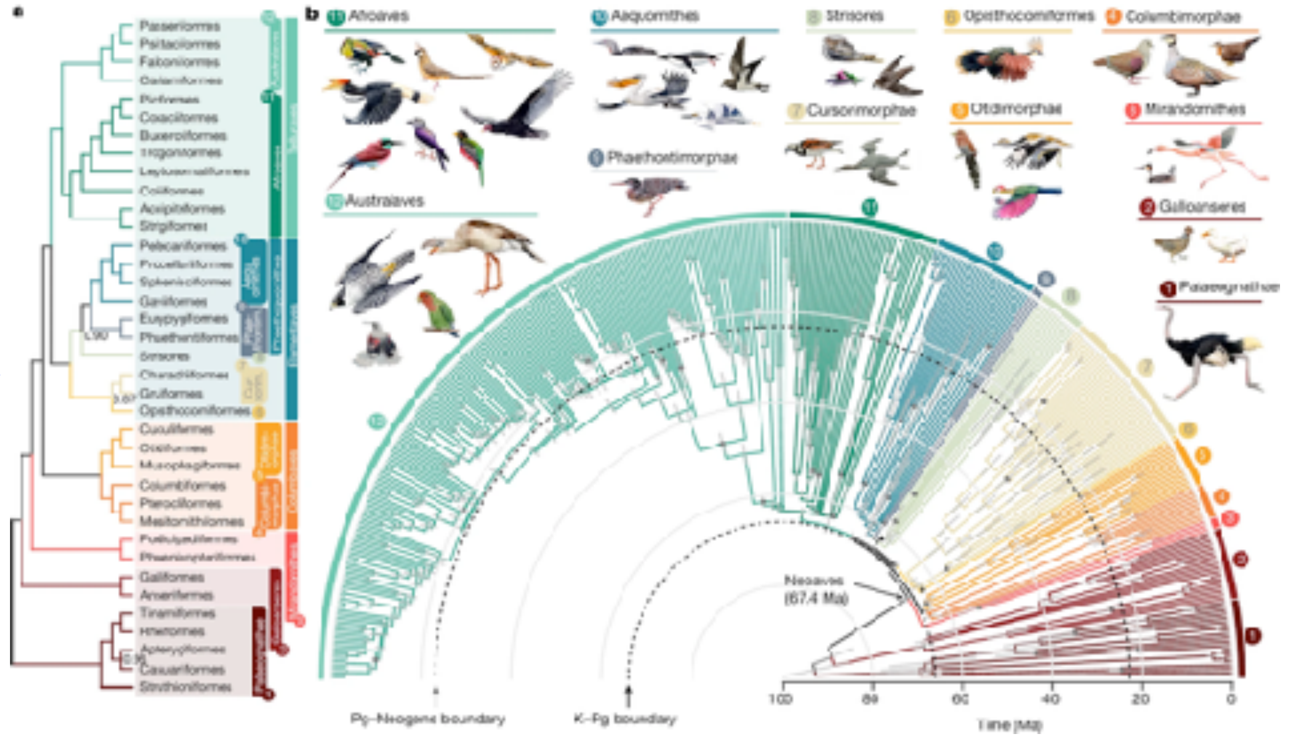
Historically biology organizes observations into useful representations and categories*:

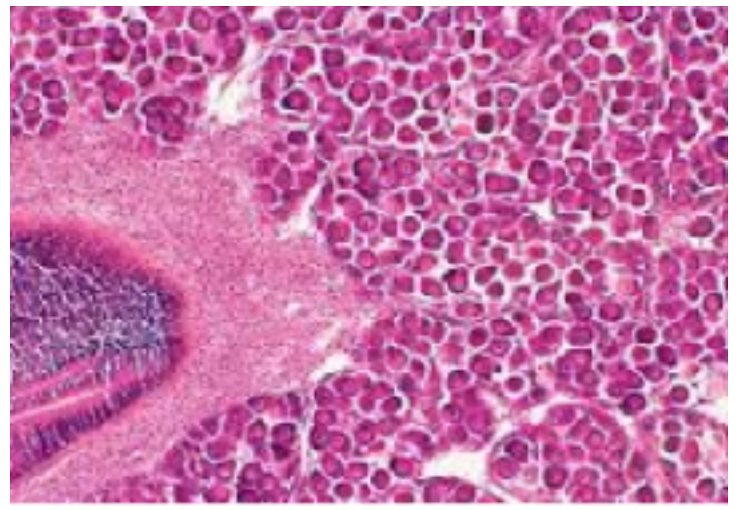
Data space X

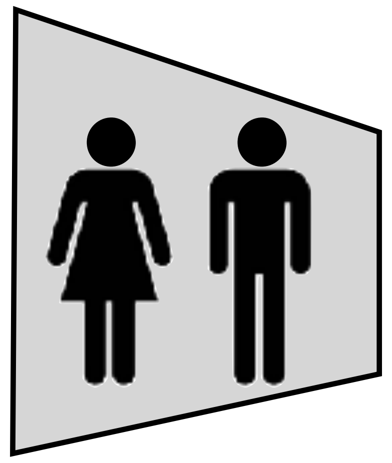
Encoder

Representation



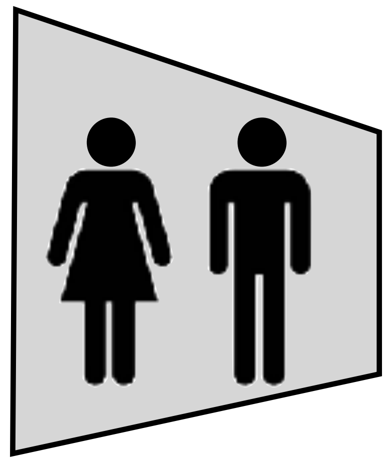






Healthy vs. **Unhealthy**

...



...

Mayr, E. (1982). The Growth of Biological Thought: Diversity, Evolution, and Inheritance

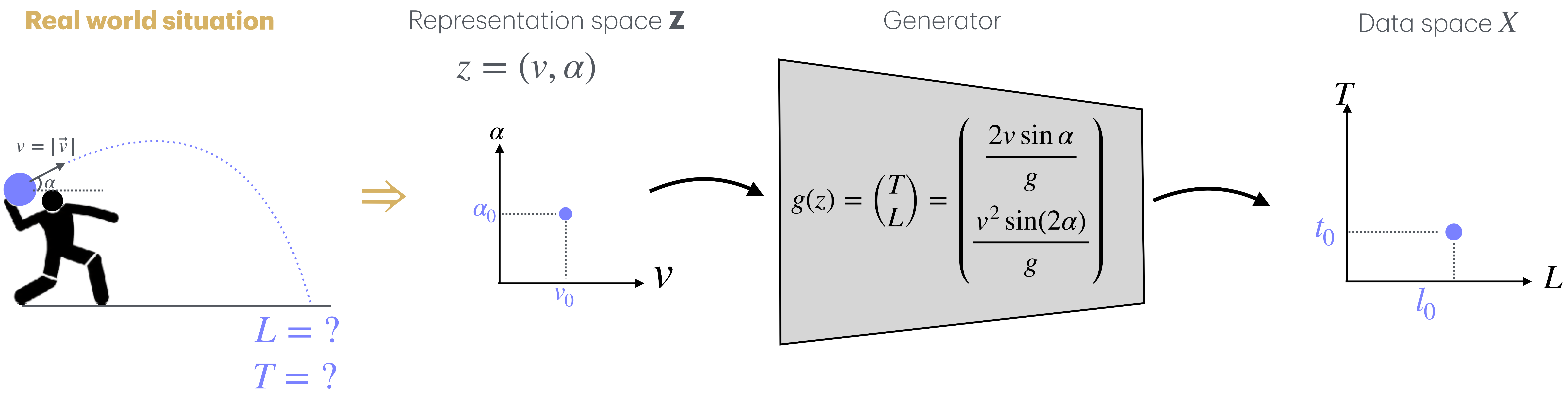
Brenner, S. (2010). Sequences and consequences.

*Many exceptions exist: Genetics, Later Developmental biology experiments, etc.

Images adapted from:
Stiller, J., Feng, S., Chowdhury, AA. *et al*. Complexity of avian evolution revealed by family-level genomes. *Nature* **629**, 851–860 (2024).
https://www.qu.edu.qa/en-us/Research/esc/Facilities/labs/Pages/histopathology.aspx

A historical intuition: physics as generative modeling

Historically, physics aims to obtain equations, symmetries, and initial conditions, then generates observable numerical predictions*.



In physics we compress the situation into a few variables, then uses laws to generate predictions.

*Caveat: this is an intuition, not a strict history. Physics also learns representations, and compact laws do not automatically explain higher-level complexity. The literature of emergence is vast and deep in physics. Eg. P. W. “More Is Different.”(1972)

A physics case study:

For pedagogy, let us take the projectile example more seriously!

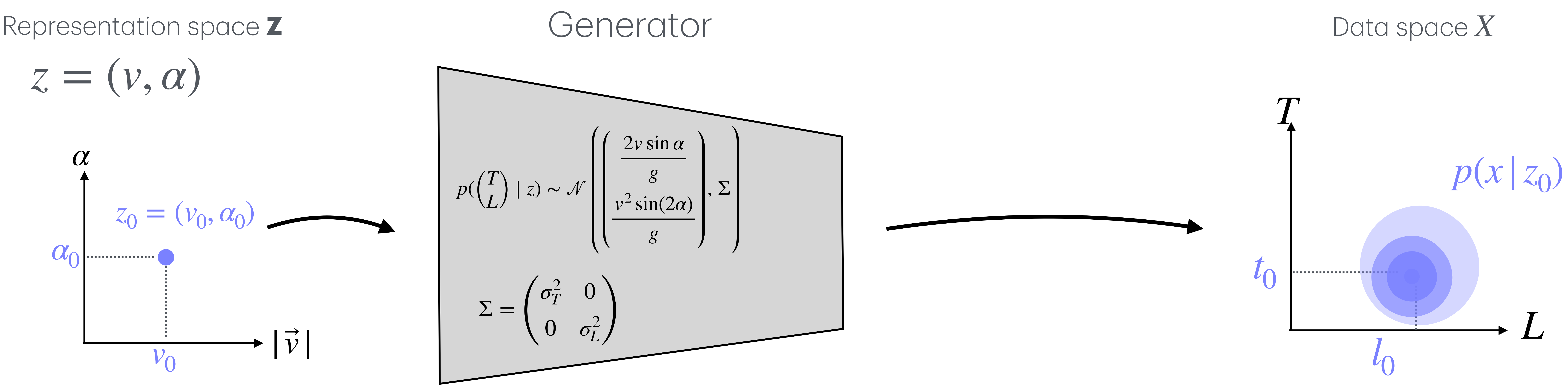
The ideal physics model says:

$$\begin{pmatrix} T \\ L \end{pmatrix} = g(z), \qquad z = (v, \alpha)$$

Idea 1: hidden/unmodeled variables: (**Wind, friction, spin, air density, etc...)**

Therefore the outputs L and T have uncertainty (randomness):

$$g(z) = \begin{pmatrix} T \\ L \end{pmatrix} \to \sim p\left(\begin{pmatrix} T \\ L \end{pmatrix} \mid z\right)$$



A physics case study:

Idea 2: We may not know the input exactly, or might want to study a variety of input values.

Instead of one point, $z_0 = (v_0, \alpha_0)$, we may only know a distribution of input values:

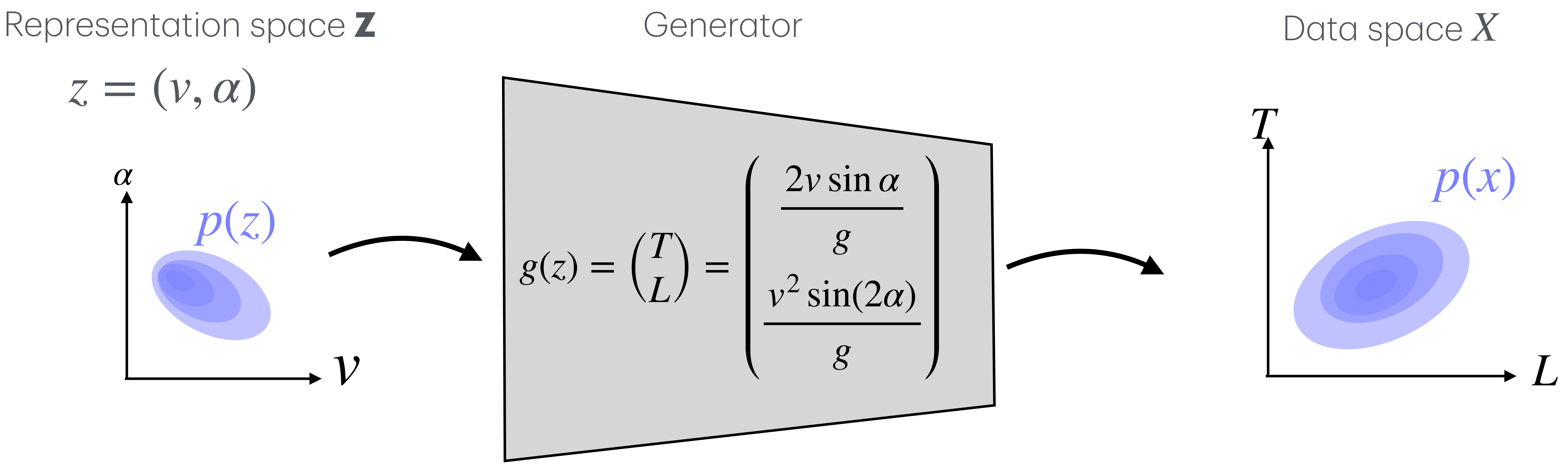
$$z = (v, \alpha) \sim p(z)$$

and then the deterministic map:

$$x = g(z), \qquad x = \begin{pmatrix} T \\ L \end{pmatrix}$$

turns that into an output distribution point by point, <u>even if the generator is deterministic</u>:

$$z \to p(z) \quad \Longrightarrow \quad x \to p(x)$$



A physics case study: Combining both ideas:

(i) **Generation**: Uncertainty and probabilistic outputs for each single point input, $\delta(z = z_0)$ (probabilistic generation):

$$p(x) = \int p(x \mid z)\delta(z - z_0)\, dz \qquad = p(x \mid z_0)$$

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Combining (i) and (ii) leads to how the output $p(x)$ is shaped based on input representation distribution $p(z)$ and generator uncertainty $p(x|z)$:

$$p(x) = \int p(x \mid z)p(z)\, dz$$

Note: *When $p(z)$ and $p(x \mid z)$ are known, $p(x)$ can be estimated by numerical integration or sampling.*

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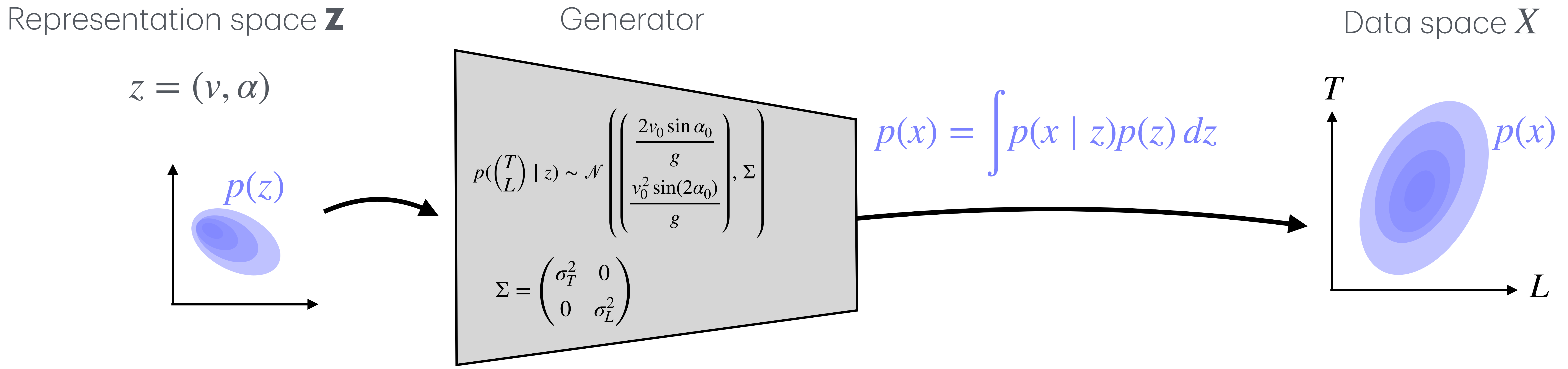


A physics case study: Combining both ideas: from deterministic maps to probabilistic generation

Let's implement this in practice:

Abstractly, we followed this direction: $p(z) \longrightarrow p(x \mid z) \longrightarrow p(x)$

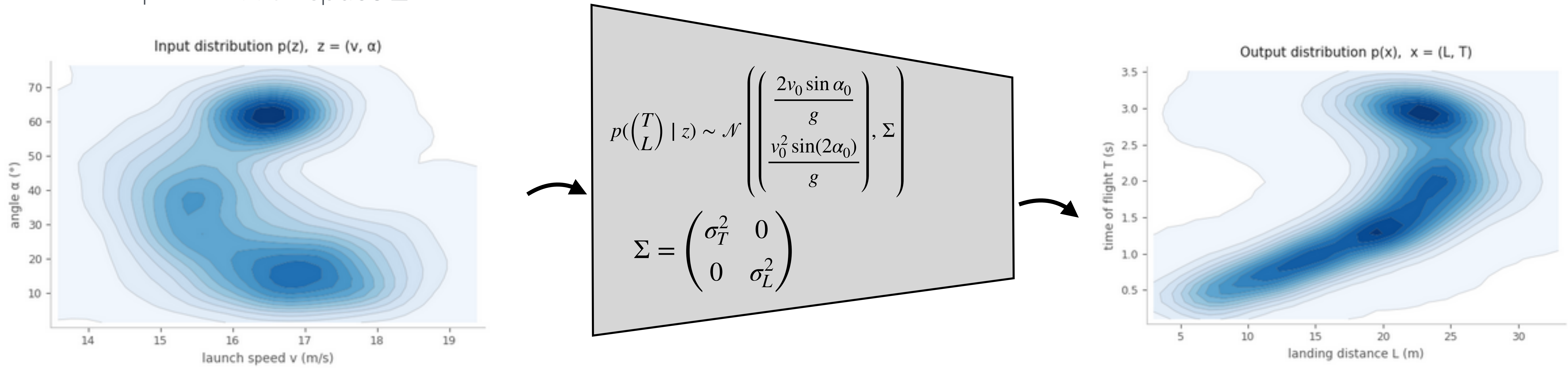
This should be given,
Or we should be able to estimate it!
We will discuss this further

This is our generator, we derived it.
Maps one input point (z) into a probability $p(x \mid z)$

This is driven by knowing the $p(z)$ and knowing the generator:

$$p(x) = \int p(x \mid z) p(z)\, dz$$

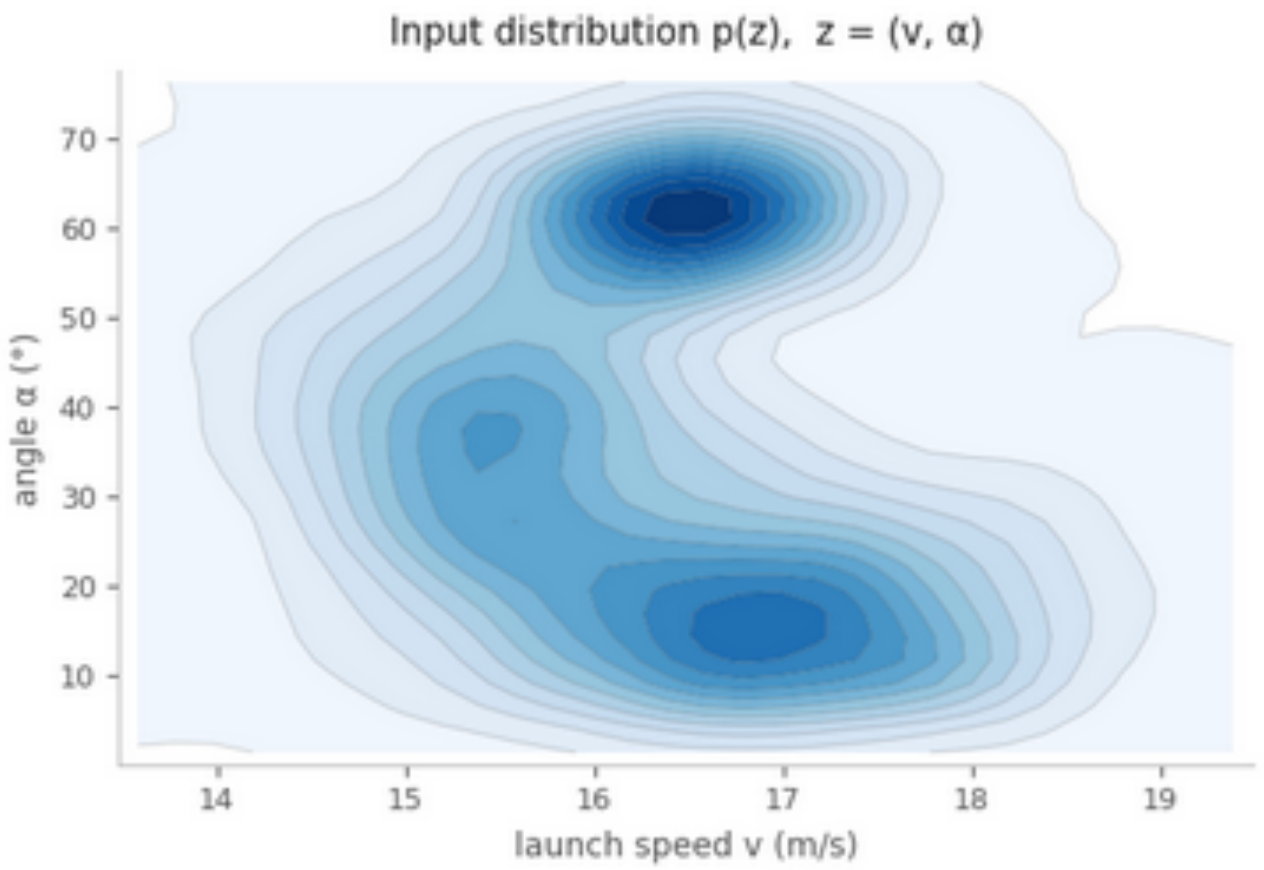
Representation space $\mathbf{z}$ | Generator | Data space X



But how did we actually dod it? We did it by sampling actual points from $p(z)$.
Let's explore this next and via this we will introduce some other important concepts of **Generative modelling**.

A physics case study: How did we approximate p(x)?

Representation space **Z**



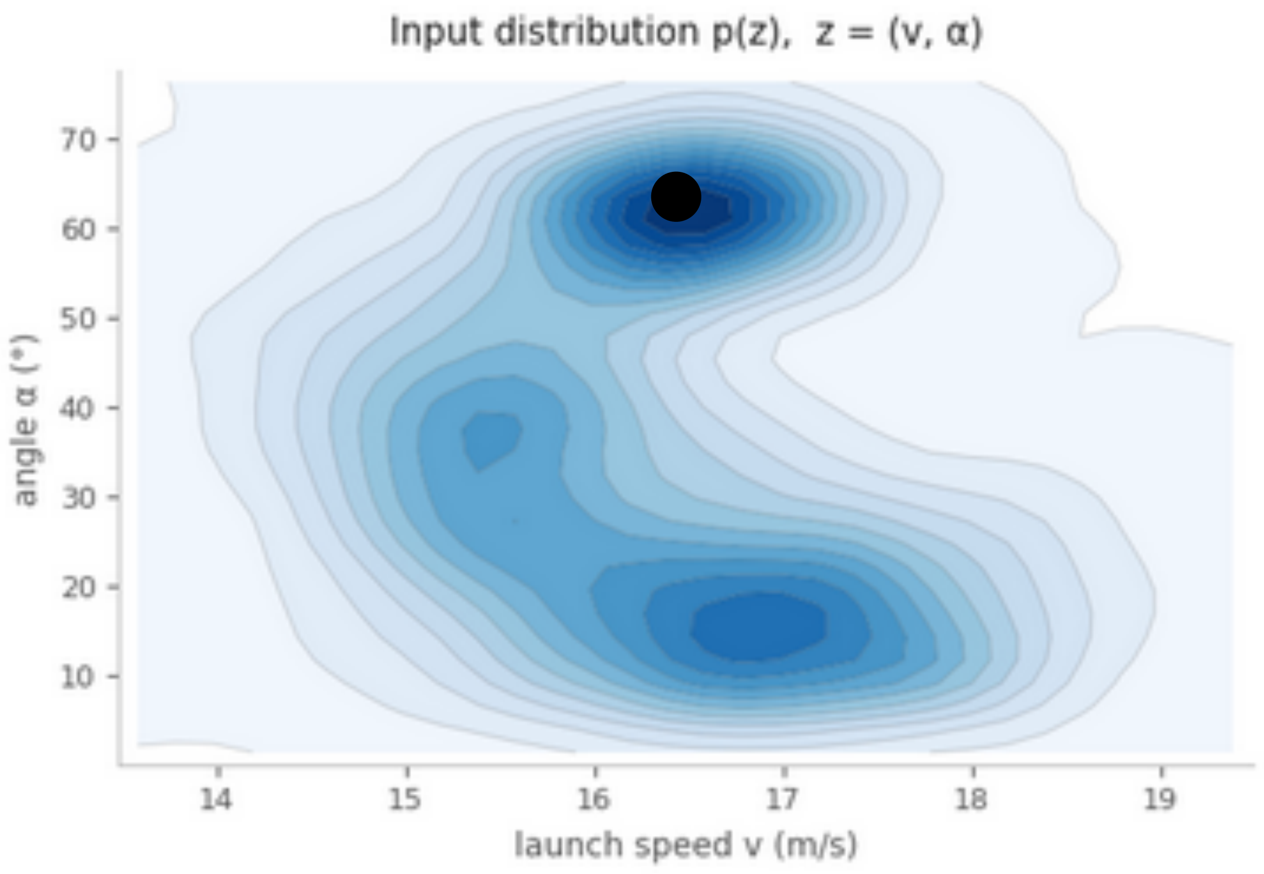
Step 1) <u>Choose an input distribution:</u>

A physics case study: How did we approximate p(x)?

Representation space **Z**

Generator

Data space X



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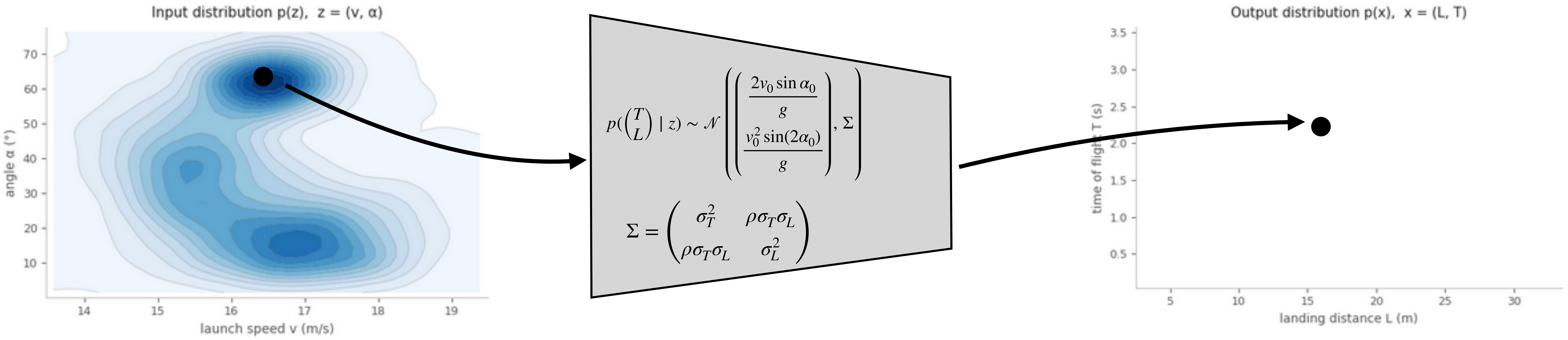
A physics case study: How did we approximate p(x)?

Representation space **Z**

Generator

$$p(x \mid z_i)$$

Data space X



$$p\left(\begin{pmatrix} T \\ L \end{pmatrix} \mid z\right) \sim \mathcal{N}\left(\begin{pmatrix} \frac{2v_0 \sin \alpha_0}{g} \\ \frac{v_0^2 \sin(2\alpha_0)}{g} \end{pmatrix}, \Sigma\right)$$

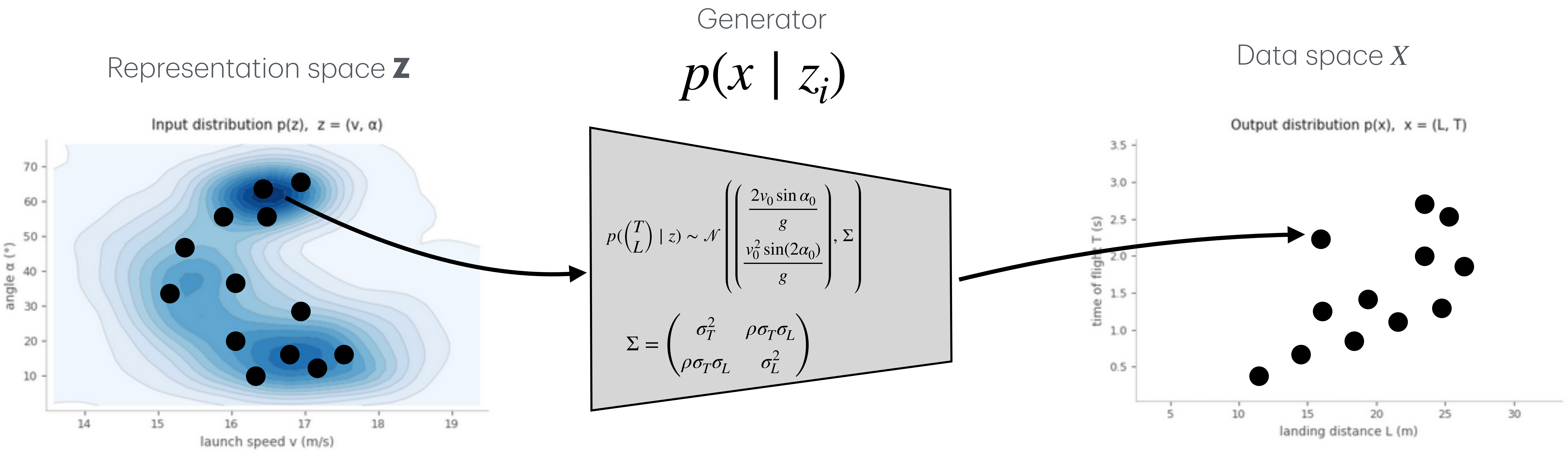
$$\Sigma = \begin{pmatrix} \sigma_T^2 & \rho\sigma_T\sigma_L \\ \rho\sigma_T\sigma_L & \sigma_L^2 \end{pmatrix}$$

Step 1) Choose an input distribution:

Step 2) Sample one point from $p(z)$

Step 3) Give that point to the generator. For this input, the generator produces one output sample: $x_i \sim p(x \mid z_i)$

A physics case study: How did we approximate p(x)?



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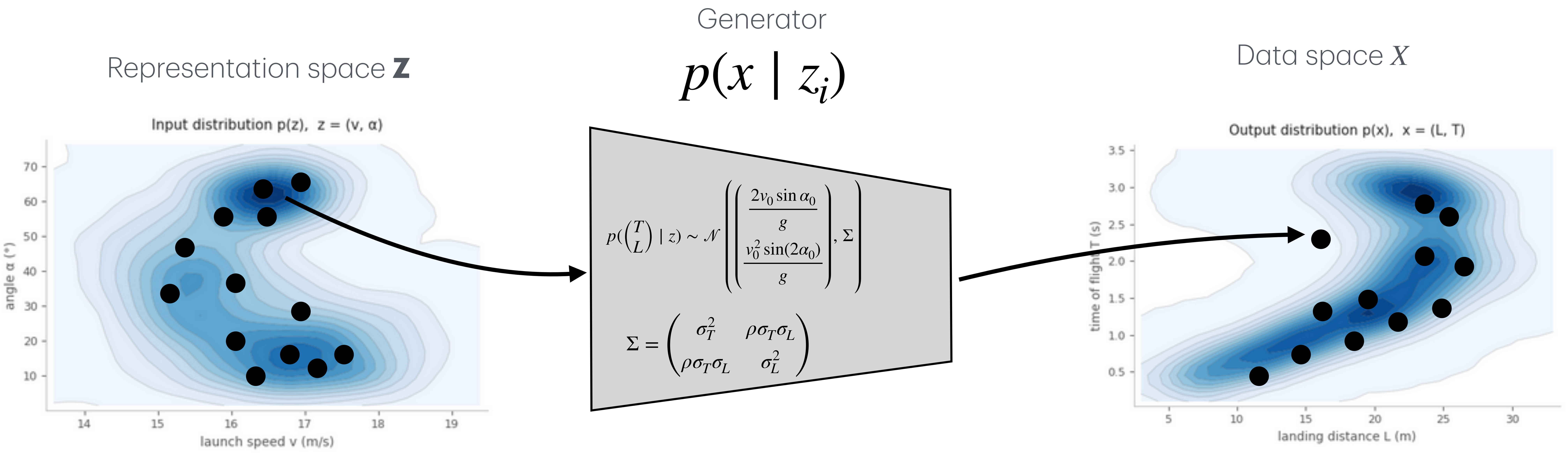

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Step 5) Estimate the output distribution: $\{x_i\}_{i=1}^{N}$ approximates $p(x)$

Bonus A physics case study: How did we approximate p(x)?

We want to compute: $p(x)$
But instead of solving the integral directly, we use sampling.

Step 1) Choose an input distribution: First, we defined a probability density in representation space: $z = (v, \alpha)$, $z \sim p(z)$
This means some launch conditions are more likely than others, and we knew it's exact form

Step 2) Sample one point from $p(z)$: We randomly choose one input point: $z_i \sim p(z)$
Points in high-density regions of $p(z)$ are more likely to be selected.

Step 3) Give that point to the generator. For this input, the generator produces one output sample: $x_i \sim p(x \mid z_i)$
The sample still has randomness because of hidden variability: wind, friction, spin, measurement noise, etc.

Step 4) Repeat many times:

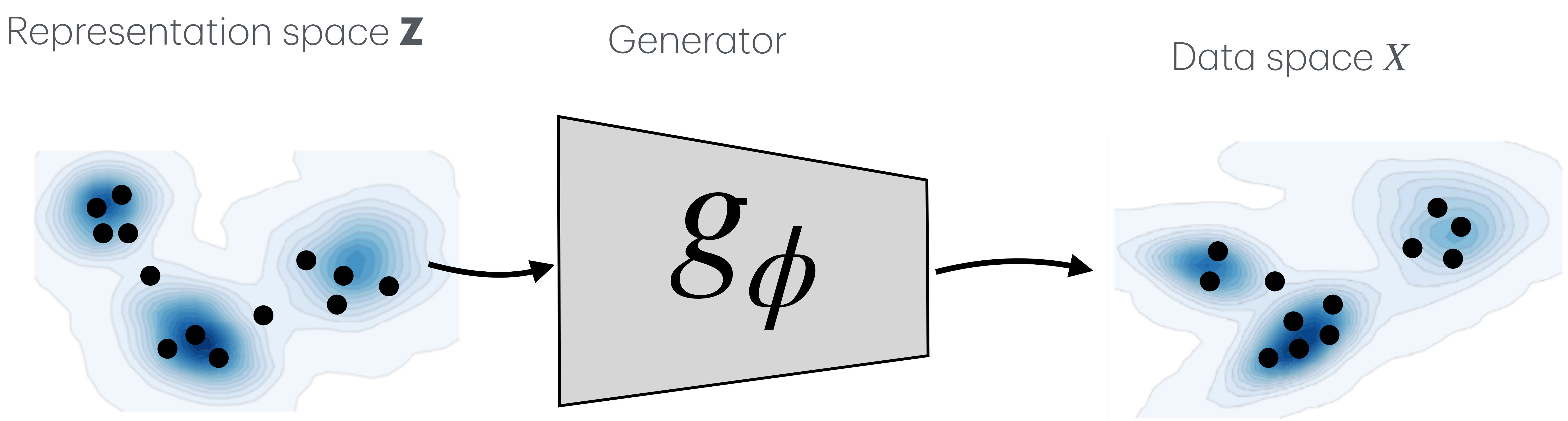
$$z_1, z_2, \ldots, z_N \sim p(z) \qquad \rightarrow \qquad x_1, x_2, \ldots, x_N, \qquad x_i \sim p(x \mid z_i)$$

These $\{x_i\}$ are samples from the output distribution $p(x)$.

Step 5) Step 5) Estimate the output distribution After many repetitions: $z_i \sim p(z)$, $x_i \sim p(x \mid z_i)$,
the generated samples are samples from the output distribution $p(x)$: $\{x_i\}_{i=1}^{N} \approx p(x)$
In another word, we can approximate $p(x) = \int p(x \mid z)p(z)\, dz$ by generating samples from $p(x)$

This is called Monte Carlo (MC) approximation, we can approximate distributions by repeated sampling.

A physics case study: The generator function can be any function g_ϕ.



Step 1) If the input distribution in the representation space $p(z)$ is known or give

Step 2) Sample one point from $p(z)$

Step 3) Give that point to the generator. For this input, the generator produces one output sample: $x_i \sim p(x \mid z_i)$

Step 4) Repeat many times

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This is a general recipe for if we **know $p(z)$, and g_ϕ already**, which is **not the case** when we move out of the domain of physics. Generative AI aims to partially address this as we will learn. Before that, we will review some basic probability language.

Background: probability

A) Probability theory: Basics

Random variables, samples, and densities:

Random variable:

$X \in \mathcal{X}$

$X \sim p(x)$: Here $p(x)$ describes how likely different values of X are.

$X_i =$ the random variable before we observe it

$x_i =$ the actual observed value after sampling

$\mathcal{D} = \{x_1, x_2, \ldots, x_N\}$ is a dataset

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Mass vs density:

For discrete data:

$p(x) = P(X = x)$ probability

$$\sum_{x \in \mathcal{X}} p(x) = 1$$

For continuous data:

$p(x) \geq 0$ probability density,

$$\int_{\mathcal{X}} p(x)\, dx = 1$$

For a region A: $P(X \in A) = \int_A p(x)\, dx$: probability

A) Probability theory: Conditional, joint, and marginal distributions

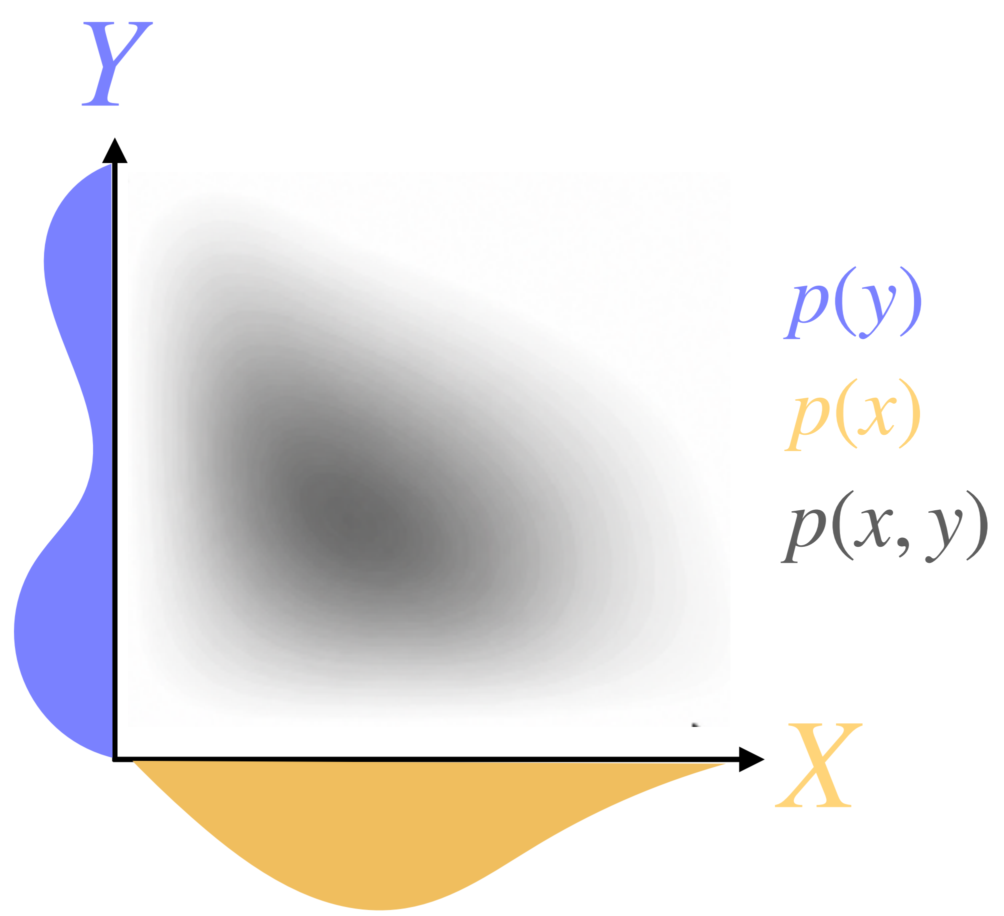
Let X and Y be two random variables. Their observed values are:

$$x \in \mathscr{X}, \qquad y \in \mathscr{Y}$$

Marginal distributions: Describe each variable alone:

Marginal distribution of X: $p(x)$

Marginal distribution of Y: $p(y)$



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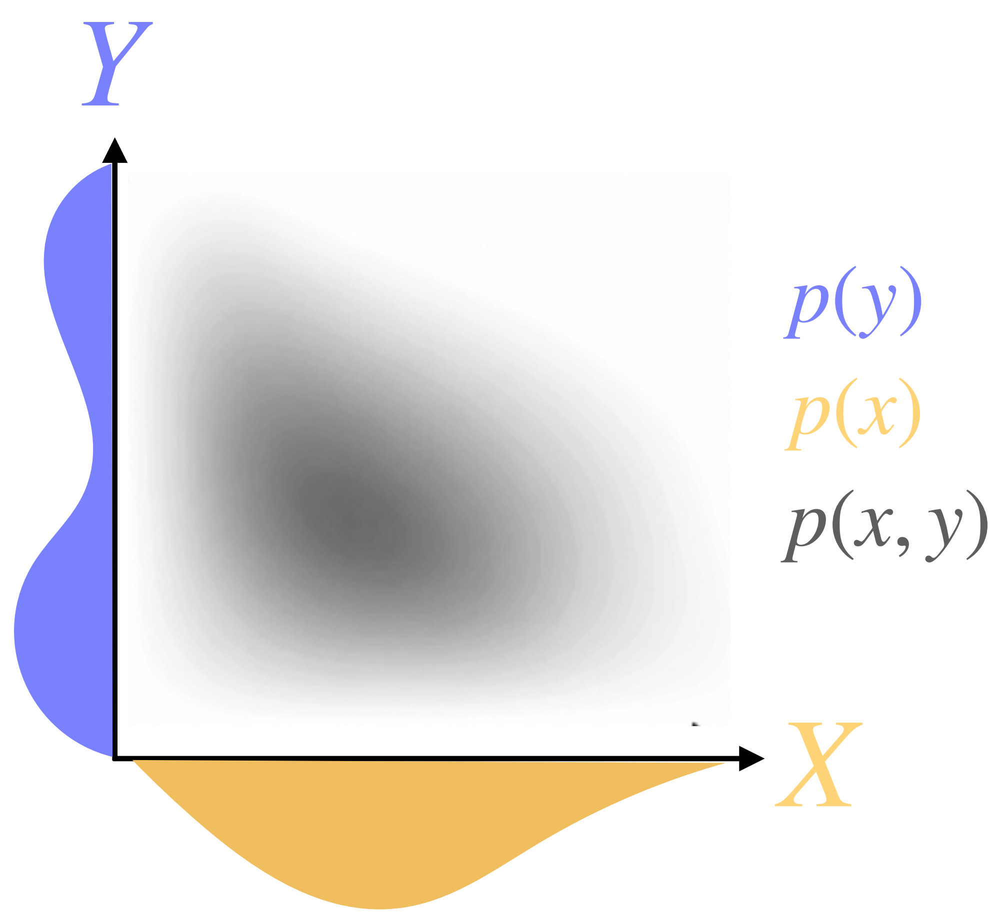

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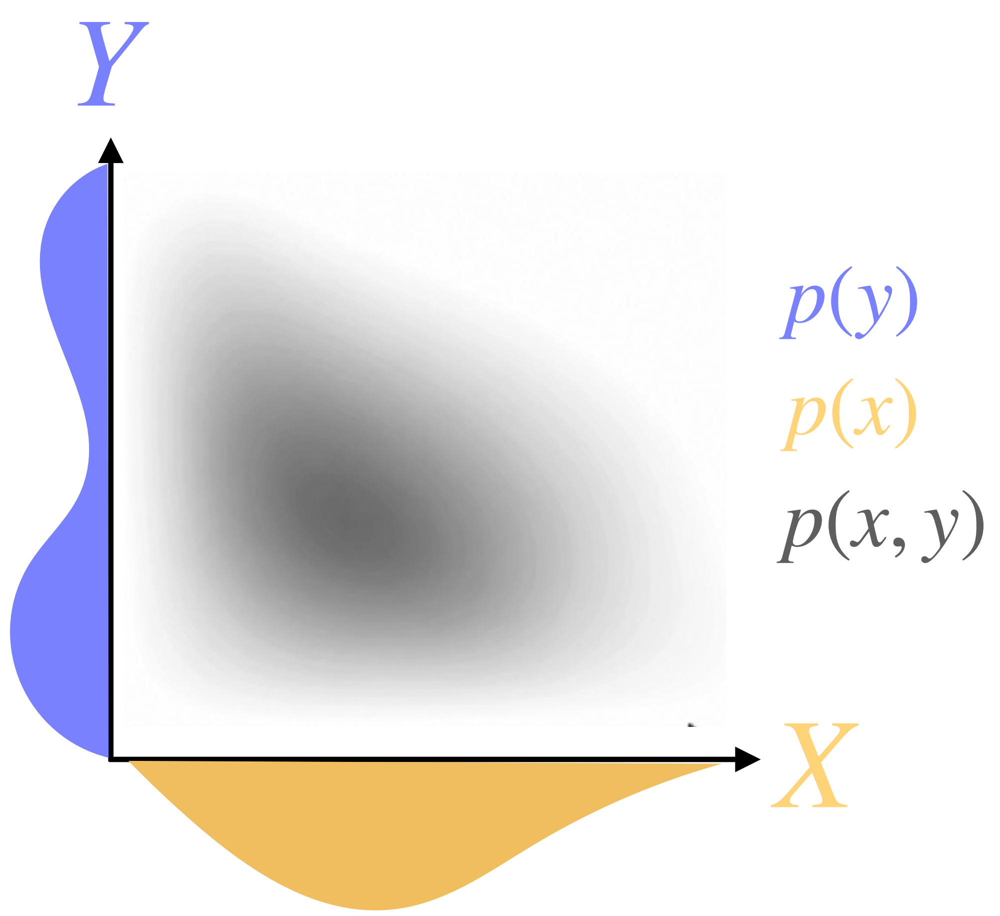

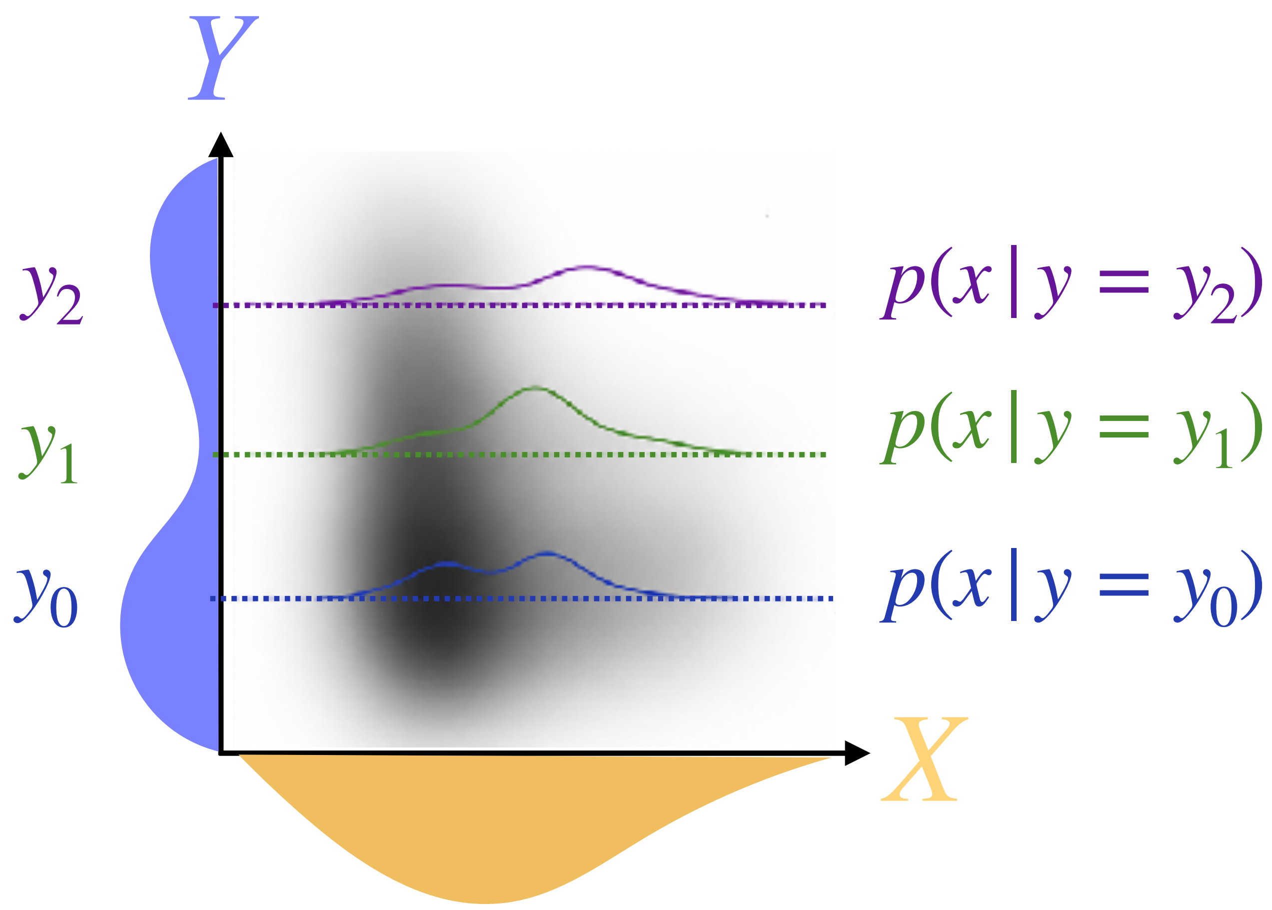

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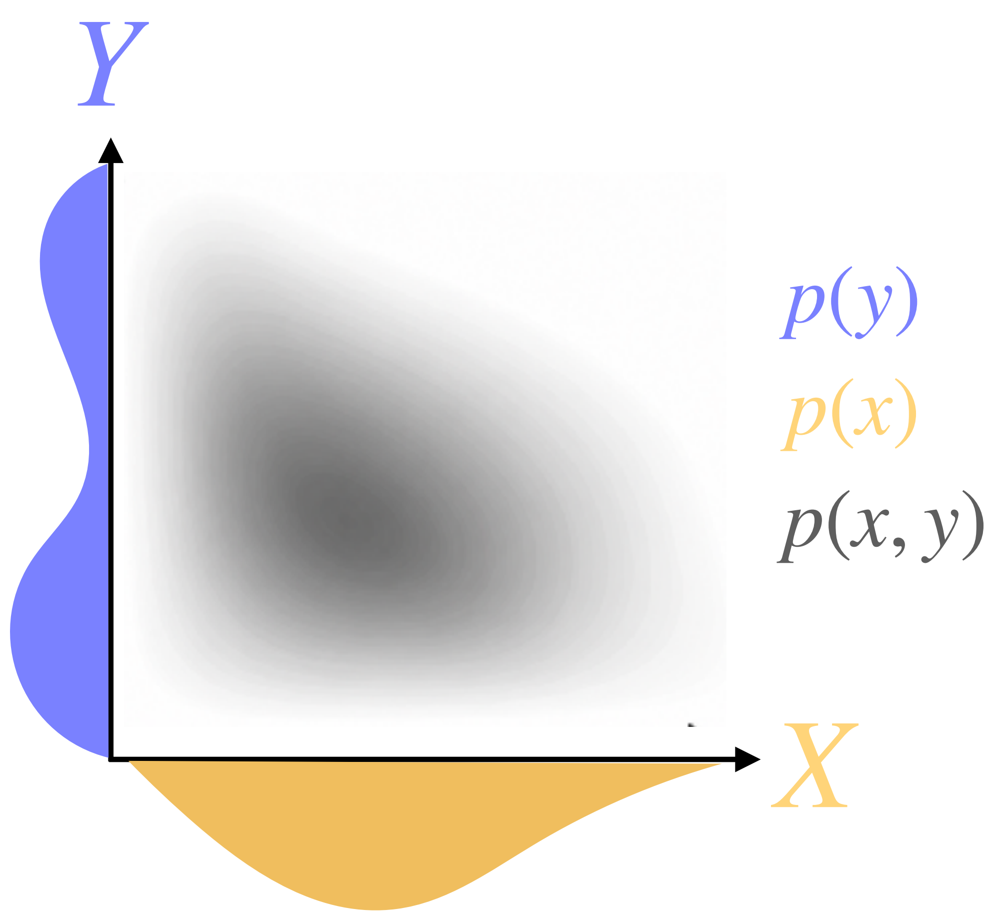
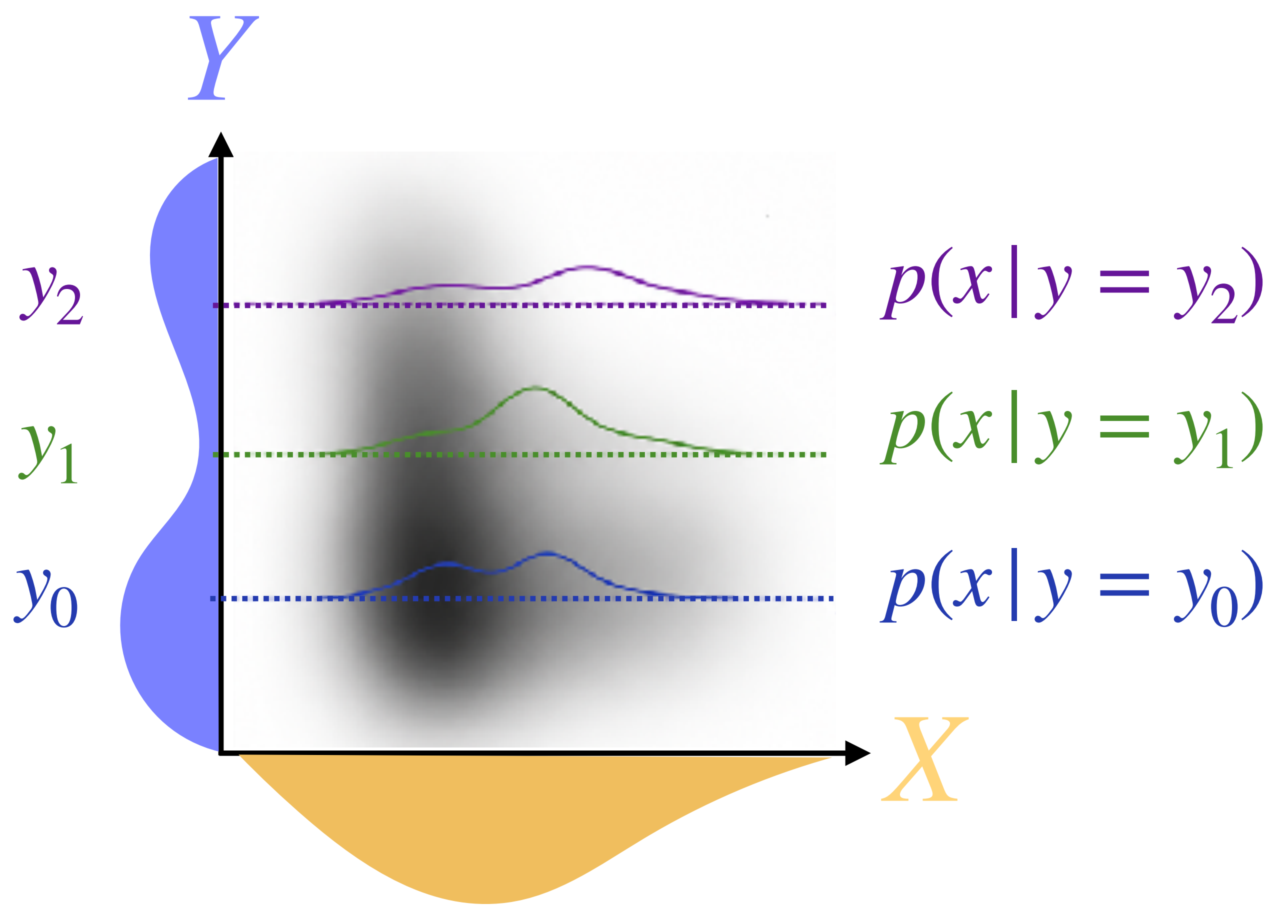
Product rule: The joint can be written as:

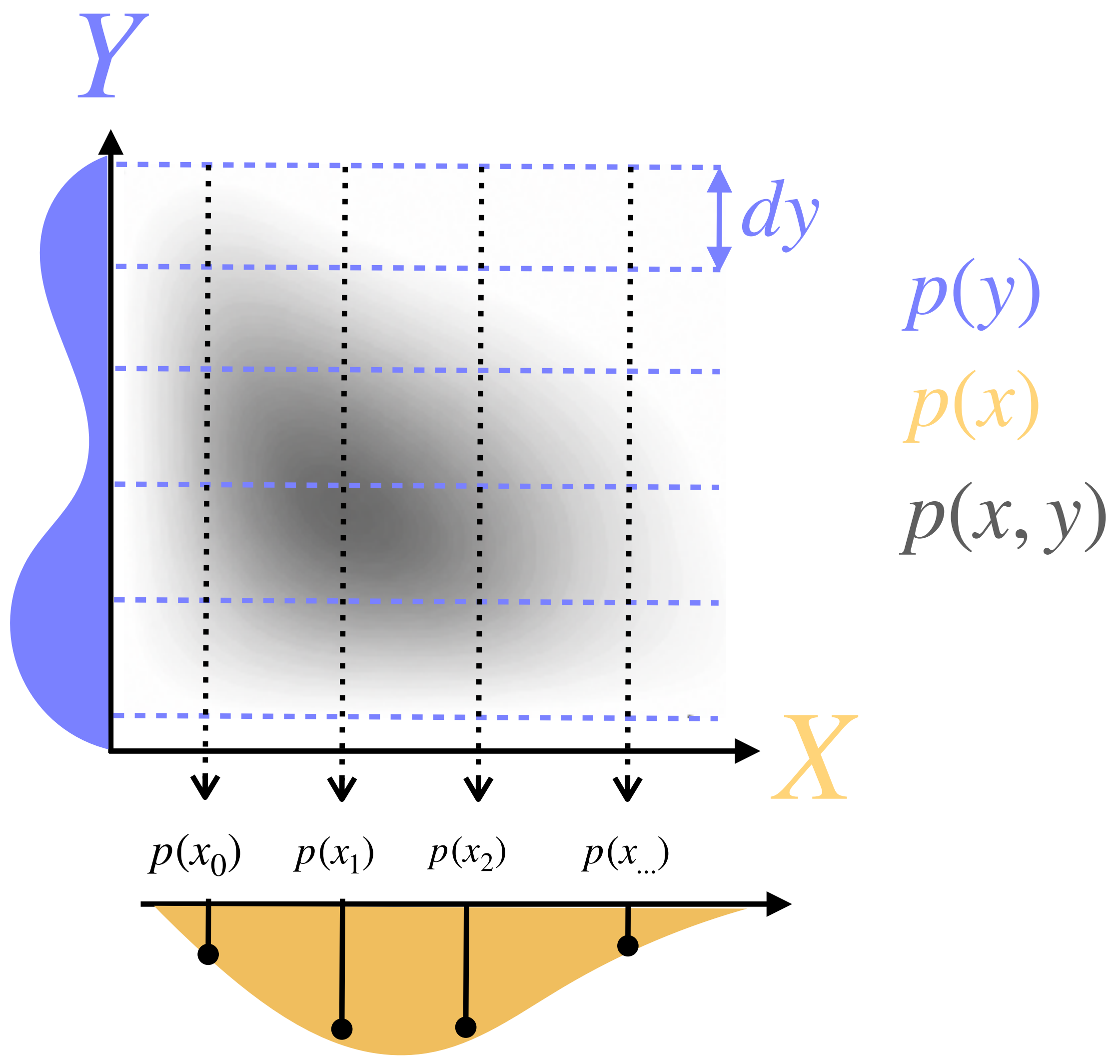
$$p(x, y) = p(x \mid y)p(y) = p(y \mid x)p(x)$$

Marginalization: To get $p(x)$, we sum/integrate over all possible y:

$$p(x) = \int p(x, y)\, dy \qquad \text{or} \rightarrow\ p(x) = \int p(x \mid y)p(y)\, dy$$







$$p(x) = \int p(x, y)\, dy$$

B) Probability application: learning a density model from "data points"

Data: Lets assume we have observed some data points:

$$\mathscr{D} = \{x_1, \ldots, x_N\}$$

We assume these are coming from a distribution that we don't have access to:

$$x_i \sim p_{data}(x) \text{ , and } p_{data} \text{ is unknown to us}$$

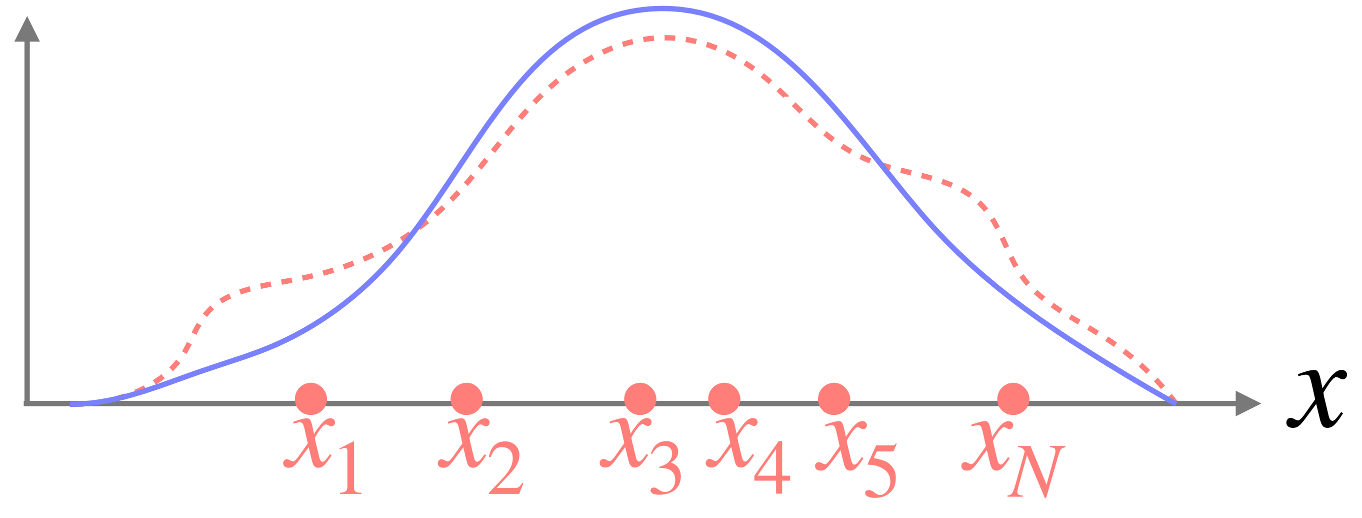
Where should we start?

Model: We can try to build a <u>*parametrized model*</u> p_θ with limited number of parameters θ to model p_{data}:

$$p_\theta(x) \approx p_{data}(x)$$

p_{data} : Unknown

p_θ : We want to estimate from $\{x_i\}$



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So we want these two be similar, but how can we define similarity of distributions?

Distance of distributions: If $p_1(x)$ and $p_2(x)$ be two probability distributions over the same domain $x \in \mathscr{X}$, KL divergence measures their distance:

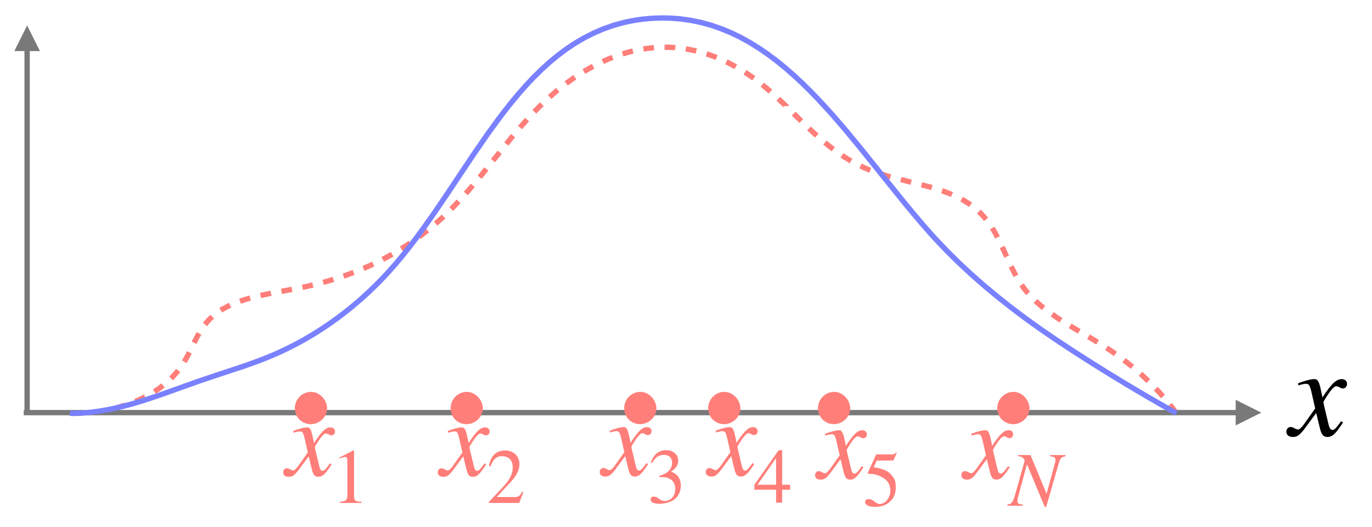
$$D_{\mathrm{KL}}(p_1 \,\|\, p_2) = \int_{\mathscr{X}} p_1(x) \log \frac{p_1(x)}{p_2(x)}\, dx \geq 0$$

Our objective: We to find model parameter θ that makes the model distribution close to the data distribution:

$$\theta^* = \arg\min_\theta D_{\mathrm{KL}}\left(p_{data} \,\|\, p_\theta\right)$$

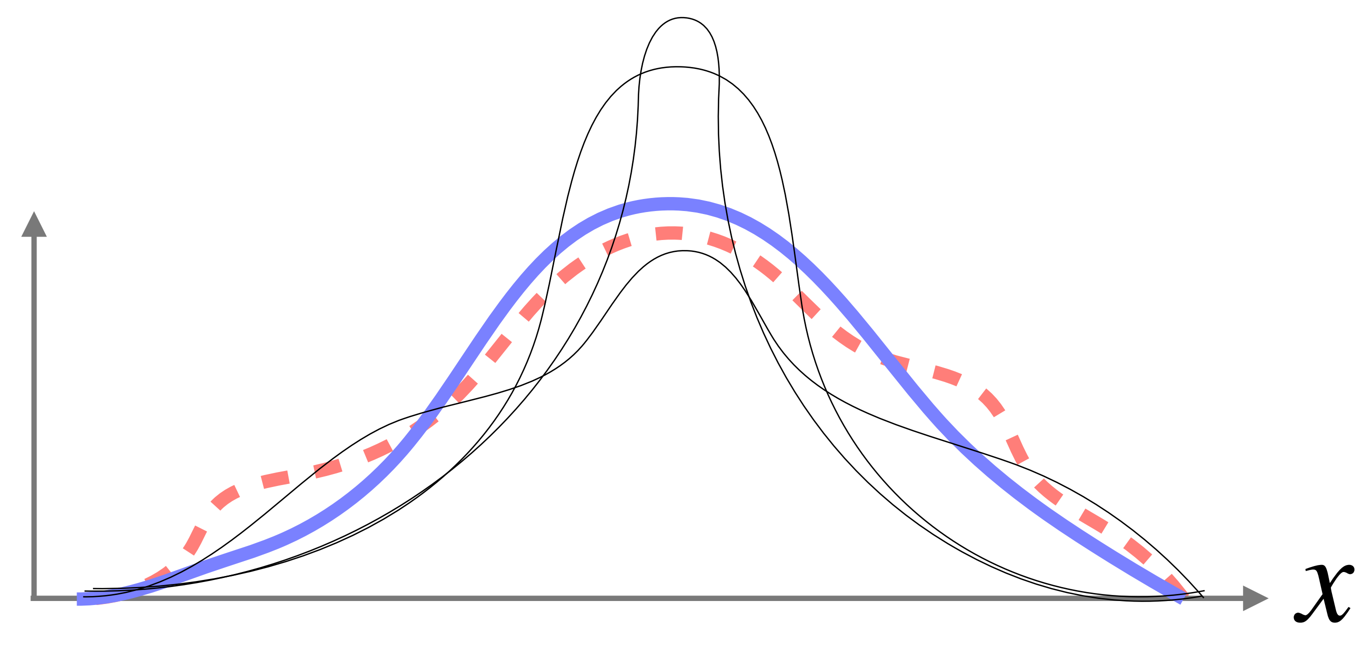
p_{data} : Unknown

p_θ : We want to estimate from $\{x_i\}$



p_θ for optimal θ^* that minimizes $D_{\mathrm{KL}}\left(p_{data} \,\|\, p_\theta\right)$

p_θ for non optimal θ



The problem is p_{data} is not known to be used in the D_{KL} formula!

Bonus B) Probability application: learning a density model from "data points"

Our objective: $\theta^* = \arg\min_{\theta} D_{\mathrm{KL}}\left(p_{data} \,\|\, p_\theta\right)$. We don't have the p_{data}, we have $\{x_i\}$.

How to solve it?

Step1: Open up the KL:

$$D_{\mathrm{KL}}\left(p_{data} \| p_\theta\right) = \int p_{data}(x) \log \frac{p_{data}(x)}{p_\theta(x)} dx = \int p_{data}(x) \log p_{data}(x) dx - \int p_{data}(x) \log p_\theta(x) dx$$

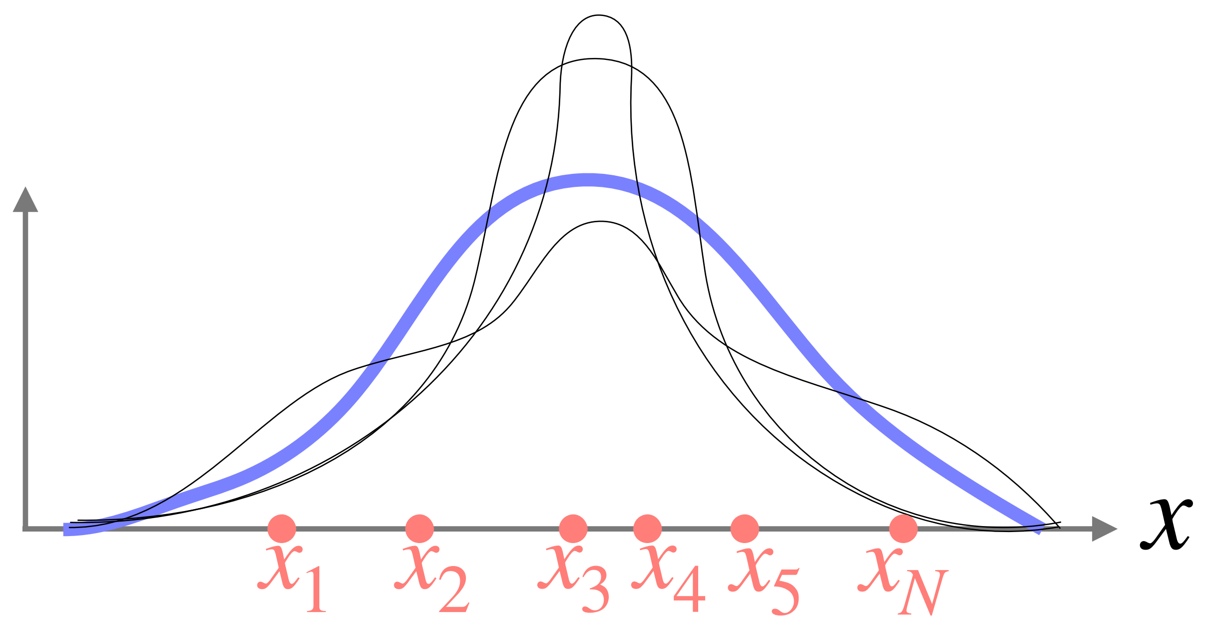
Step2: The first term does not depend on θ:

$$\theta^* = \arg\min_{\theta} D_{\mathrm{KL}}\left(p_{data} \| p_\theta\right) = \arg\max_{\theta} \int p_{data}(x) \log p_\theta(x) dx$$

Step3: Replace the integral with sample average (remember MC):

$$\theta^* = \arg\max_{\theta} \int p_{data}(x) \log p_\theta(x) dx \; = \arg\max_{\theta} \mathbb{E}_{x \sim p_{data}}[\log p_\theta(x)] \approx \arg\max_{\theta} \frac{1}{N} \sum_{i=1}^{N} \log p_\theta(x_i)$$

Now we can conclude it for: A given set of data points $\{x_i\}$ + A parameterized function $p_\theta(x)$

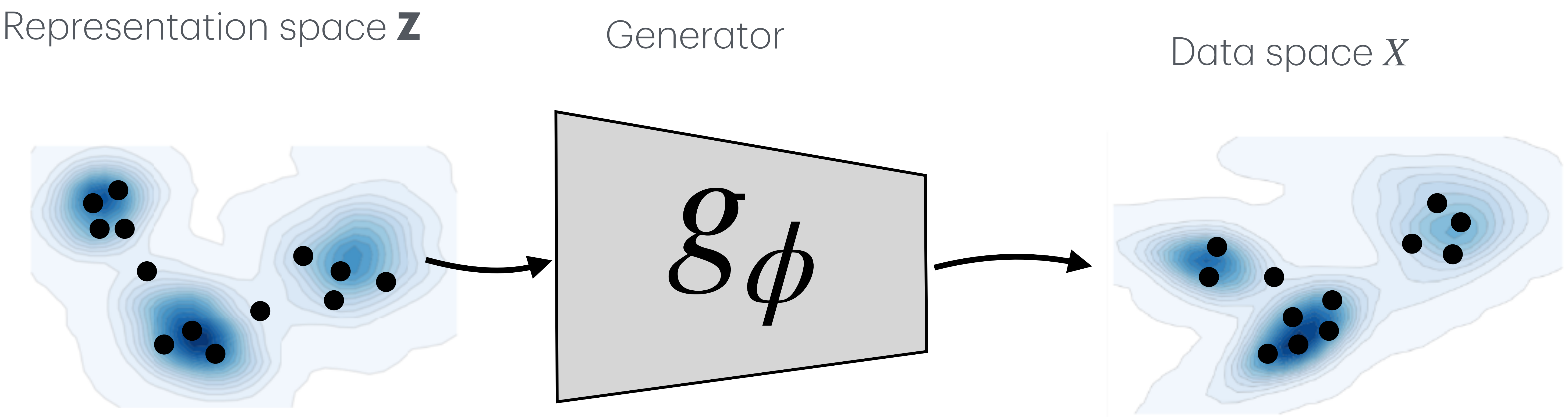


Note: if we define the likelihood of the dataset as: $\mathcal{L}(\theta) = \prod_{i=1}^{N} p_\theta(x_i)$, then the log-likelihood is:

$$\log \mathcal{L}(\theta) = \sum_{i=1}^{N} \log p_\theta(x_i)\,.$$

And the whole process above (steps 1-3) is called maximizing log-likelihood of the data.

In the next lecture we will explore this problem:



We don't know $p(z)$

We don't know g_ϕ

We don't know p_{data}

All we have is some samples in Data space $\{x_i\} \in X$